Ex:No: 1A Date: 21/1/26	Billing System Calculation Using Arithmetic Expressions

Problem Description:

A **store owner** wants to calculate the **total cost, discount, and final bill** for a customer. The store offers a **10% discount** if the total purchase exceeds 500 units. Write an R program that takes the **price of an item** and **quantity purchased** as input and calculates:

1. Total cost
2. Discount (if applicable)
3. Final bill amount.

Objective:

To calculate the total cost, applicable discount, and final bill amount using arithmetic expressions and conditional statements in R, demonstrating how decision-making logic can be applied in a real-world billing system scenario.

Sample Input	Expected Output
Enter price of item: 250 Enter quantity purchased: 3	Total cost: 750 Discount: 75 Final bill amount: 675
Enter price of item: 100 Enter quantity purchased: 5	Total cost: 500 Discount: 0 Final bill amount: 500

Program:

```
# Billing System Calculation in R
# User input price and quantity
price <- as.numeric(readline("Enter price of item: "))
quantity <- as.numeric(readline("Enter quantity purchased: "))

# Calculate total cost
total_cost <- price * quantity

# Calculate discount
if (total_cost > 500) {
    discount <- total_cost * 0.10
} else {
    discount <- 0
}
```

SAMPLE INPUT:

1) Enter price of item: 150
Enter quantity purchased: 4

EXPECTED OUTPUT:

Total cost: 600
Discount: 60
Final bill amount: 540

2) Enter price of item: 80
Enter quantity purchased: 6

Total cost: 480
Discount: 0
Final bill amount: 480

Calculate final bill

```
final_bill <- total_cost - discount
```

Display results

```
cat ("\n Total cost: ", total_cost)
cat ("\n Discount: ", discount)
cat ("\n Final bill amount: ", final_bill)
```

Result:

Thus, the program correctly calculates the total cost, discount, and final bill using arithmetic operations